A PROJECT TITLE

ON

Product Review Sentiment Analysis Using NLP and BERT

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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1. Abstract:

Product Review Sentiment Analysis is an NLP-based project designed to automatically identify and classify customer opinions expressed in product reviews. With the rapid growth of e-commerce platforms, a large number of customer reviews are generated every day, making manual analysis time-consuming and difficult. This project aims to develop an accurate sentiment classification system using Natural Language Processing (NLP) techniques and Bidirectional Encoder Representations from Transformers (BERT). The Amazon Product Reviews dataset will be used for training and testing the proposed model. The dataset contains customer reviews along with sentiment-related information, which can be used to classify reviews into categories such as positive and negative. Initially, the review data will be preprocessed by handling missing values, removing unnecessary characters, and preparing the text in a suitable format for the BERT model. Unlike traditional machine learning methods that mainly depend on individual words or word frequencies, BERT understands the context and relationships between words in a sentence, allowing it to identify sentiment more effectively. The preprocessed dataset will be divided into training and testing sets, and the BERT model will be fine-tuned using the training data. After training, the model will be tested on unseen product reviews to determine its ability to correctly classify customer sentiment. The performance of the proposed system will be evaluated using Accuracy, Precision, Recall, and F1-Score, which will provide a comprehensive measurement of the model's classification performance. A confusion matrix may also be used to visualize the correctly and incorrectly classified reviews. The developed system can help businesses understand customer satisfaction, identify common complaints, monitor product feedback, and support better decision-making. Overall, this project demonstrates how NLP and BERT can be effectively combined to build an intelligent and context-aware product review sentiment analysis system for e-commerce applications.

2. INTRODUCTION:

The rapid growth of e-commerce has made online product reviews an important source of information for both customers and businesses. Customers often depend on reviews to decide whether a product is worth purchasing, while businesses use feedback to understand customer satisfaction and improve their products and services. However, the increasing number of reviews makes it difficult to manually read and analyze every review. Reviews may also contain different expressions, opinions, and contextual meanings, making automatic analysis challenging. This motivated the development of a Product Review Sentiment Analysis system that can automatically analyze customer feedback and identify whether a review expresses a positive or negative opinion. Such a system can save time, provide quick insights, and help businesses make better data-driven decisions.

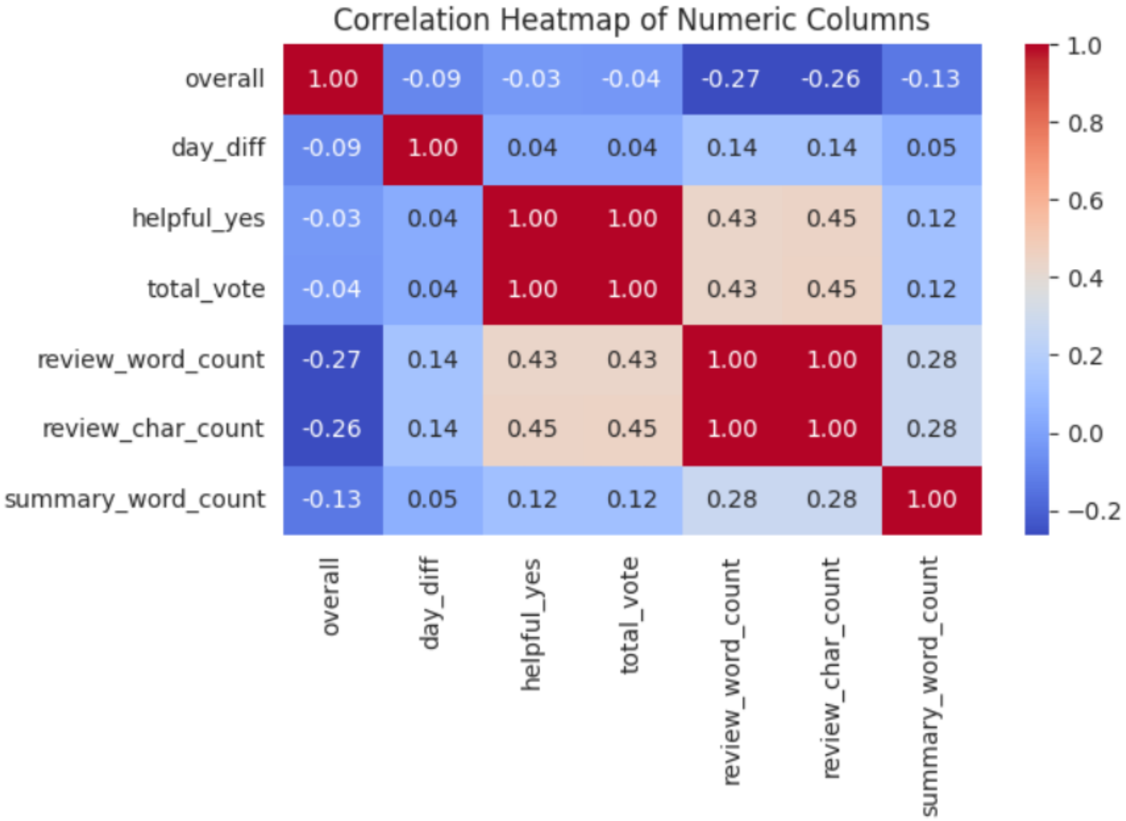
Product Review Sentiment Analysis is a Natural Language Processing (NLP) application that focuses on extracting and classifying the sentiment expressed in textual reviews. In this project, the Amazon Product Reviews dataset will be used to train and evaluate the model. The reviews will be preprocessed and then analyzed using Bidirectional Encoder Representations from Transformers (BERT), a transformer-based deep learning model capable of understanding the context and relationships between words. BERT will be fine-tuned on the product review dataset to classify reviews into sentiment categories such as positive and negative. The performance of the model will be evaluated using Accuracy, Precision, Recall, and F1-Score, along with a confusion matrix to analyze classification results. The proposed system aims to provide an efficient and reliable approach for understanding customer opinions and demonstrating the effectiveness of modern NLP techniques in sentiment analysis.

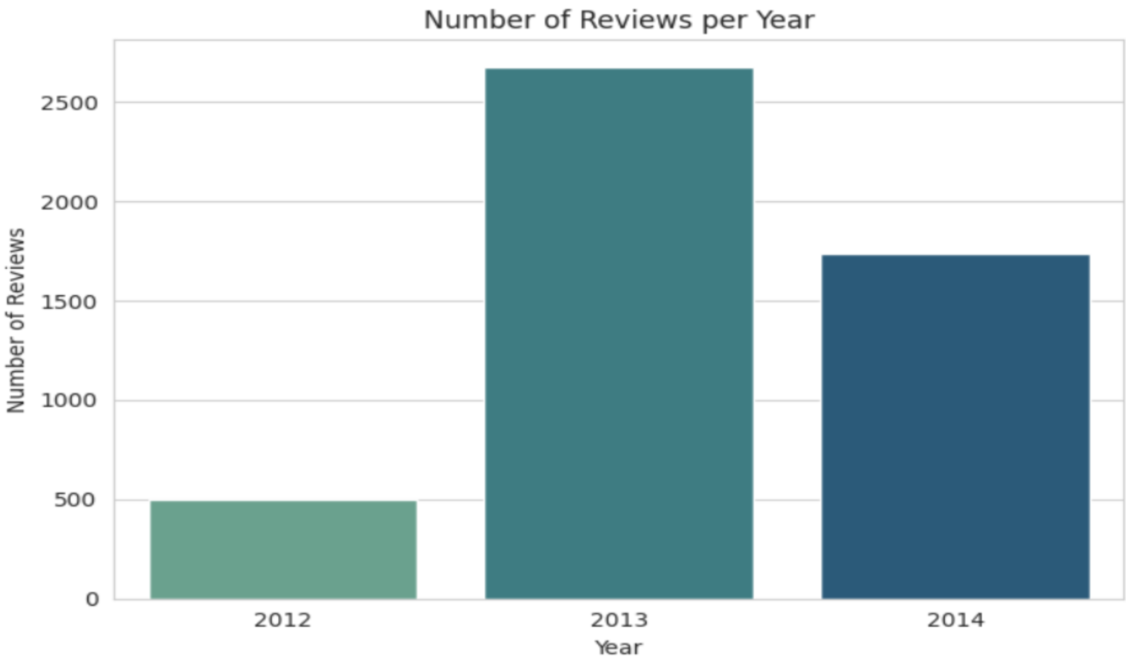
3. SOFTWARE REQUIREMENTS:

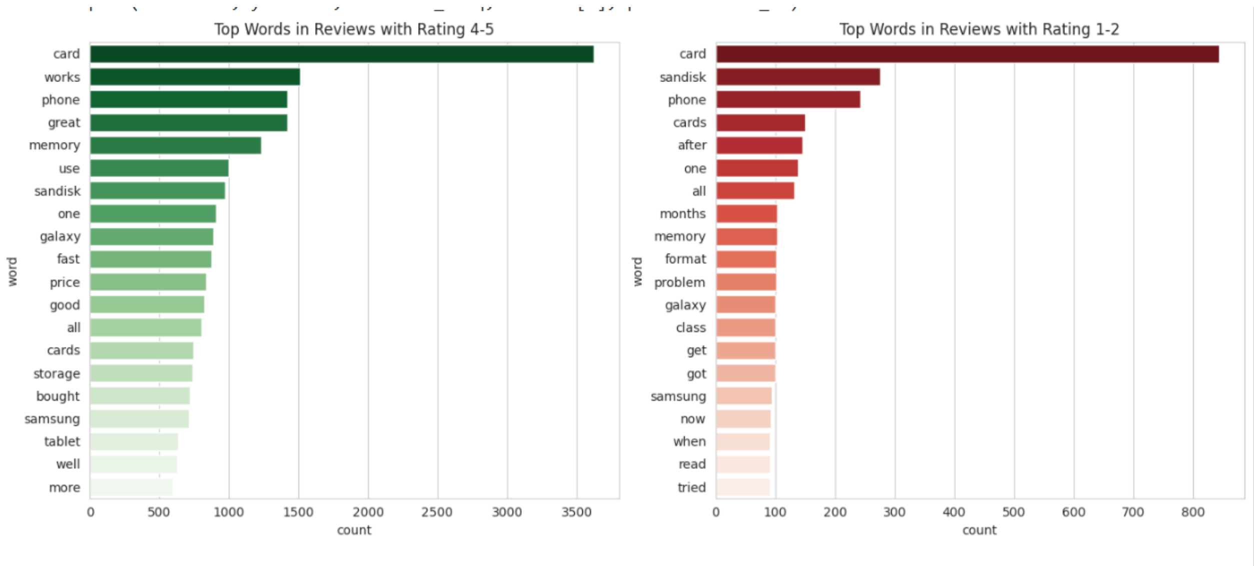
The proposed Product Review Sentiment Analysis Using NLP and BERT system requires suitable software and hardware resources for data preprocessing, model training, testing, and evaluation. The project can be developed using Python with libraries such as PyTorch/TensorFlow, Transformers, Pandas, NumPy, Scikit-learn, and Matplotlib. A development environment such as Jupyter Notebook, Google Colab, or VS Code can be used. Since BERT is a deep learning model, a system with sufficient RAM and a GPU is recommended for faster model training. However, the project can also be implemented using a CPU for smaller datasets.

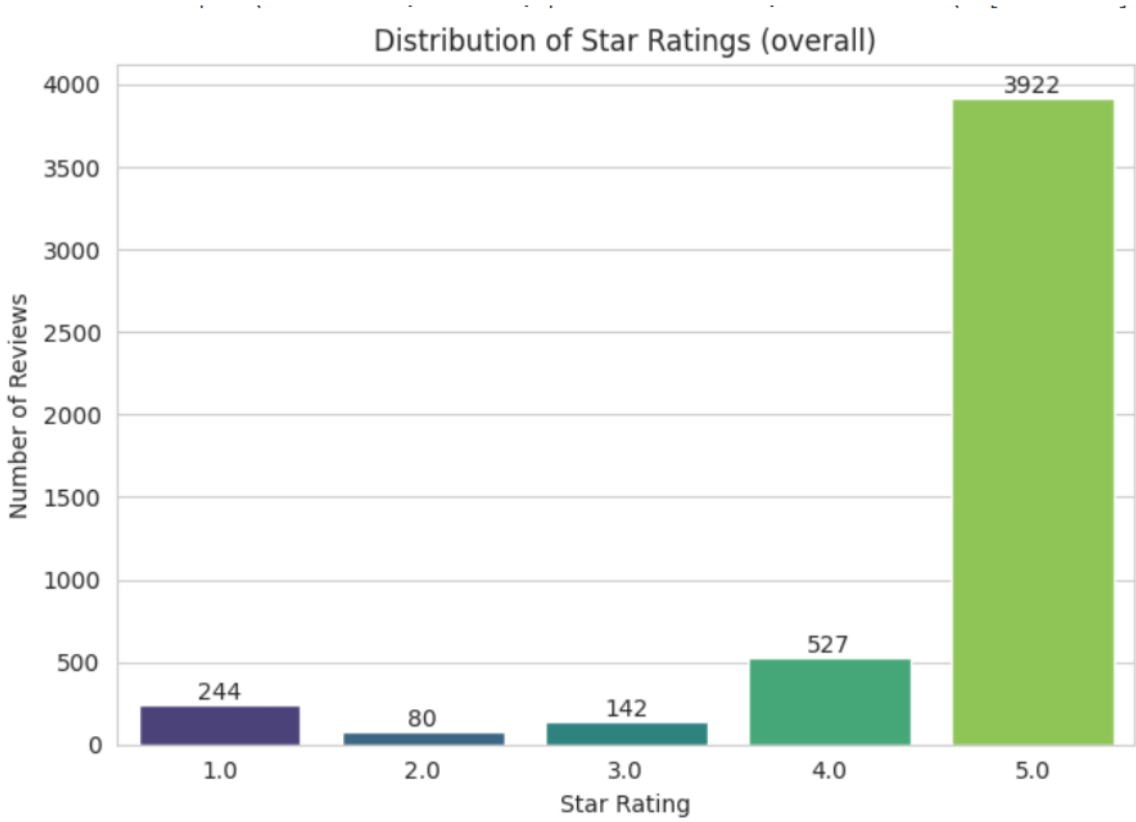
Category	**Requirement**	**Specification**
Hardware	Processor	Intel Core i5 or equivalent
	RAM	Minimum 8 GB, Recommended 16 GB
	Storage	Minimum 10 GB free space
	GPU	NVIDIA GPU with CUDA support
	Internet	Required for downloading datasets and BERT models
Software	Operating System	Windows / Linux / macOS
	Programming Language	Python 3.x
	IDE/Environment	Jupyter Notebook / Google Colab / VS Code
	Deep Learning Framwork	PyTorch or TensorFlow
	NLP Library	Hugging Face Transformers
	Data Processing	Pandas, NumPy
	Machine Learning	Scikit-learn
	Visualization	Matplotlib
	Dataset	Amazon Product Reviews Dataset

EDA Outputs:


Correlation Heatmap of Numeric Columns
overall 1.00 -0.09 -0.03 -0.04 -0.27 -0.26 -0.13
day_diff -0.09 1.00 0.04 0.04 0.14 0.14 0.05
helpful_yes -0.03 0.04 1.00 1.00 0.43 0.45 0.12
total_vote -0.04 0.04 1.00 1.00 0.43 0.45 0.12
review_word_count -0.27 0.14 0.43 0.43 1.00 1.00 0.28
review_char_count -0.26 0.14 0.45 0.45 1.00 1.00 0.28
summary_word_count -0.13 0.05 0.12 0.12 0.28 0.28 1.00
overall
day_diff
helpful_yes
total_vote
review_word_count
review_char_count
summary_word_count
1.0
0.8
0.6
0.4
0.2
0.0
−0.2


Number of Reviews per Year
Number of Reviews
2500
2000
1500
1000
500
0
2012
2013
2014
Year





Signature of the Guide **Course Instructor**